

UCC402: GEOINFORMATICS

L	T	P	Cr
3	0	3	4.5

Course Objective: Surveying as a subject in civil engineering aims to provide basic knowledge about the principles of surveying for a location and its application in the design and construction of engineering projects. The students develop skills using surveying instruments, including measuring tapes, theodolites, and advanced measurement equipment, such as laboratory stations, GPS, and drones.

Introduction to Surveying: Definition, classification of surveys, Principle, distorted or shrunk scales, precision in surveying, different types of errors.

Linear Measurements:

- (a) Chain and Tape Surveying: Chain surveying, tapes, ranging–direct & indirect, chaining on sloping ground, corrections for linear measurements, offsets
- (b) Leveling: Definitions of terms used in leveling, different types of levels, temporary adjustments, benchmarks, booking and reducing the levels, rise and fall method, line of collimation method, errors in leveling, permanent adjustments, corrections to curvature and refraction, setting out grades.
- (c) Tacheometry: Definitions and terms used in tacheometry, determination of constants, angular tacheometry with staff vertical and staff inclined, Merits and Demerits; Analytic lens, tangential method of tachometry.
- (d) Contours: Definition, representation of reliefs, horizontal equivalent, contour interval, characteristics of contours, methods of contouring, contour gradient, uses of contour maps.

Angular Measurements:

- (a) Compass Traversing: Types of compasses, bearings, meridians, declination, the dip of the magnetic needle, bearing of lines from included angles, local attraction.
- (b) Theodolite: Types of theodolites, measurement of angles, temporary and permanent adjustments, closed and open traverse, consecutive and independent coordinates, advantages and disadvantages of traversing closing error, Gale's traverse table.

Curve Setting: Types of curves, calculations for simple circular curve and tangential curve, Rankine's method of tangential angle measurements.

Total Station: Working principle and survey with total station

Global Positioning Systems (GPS): Working Principle, Types of GPS, Application of GPS, DGPS-working Principle, DGPS errors (RT & PP).

Remote Sensing and GIS: Introduction to Remote sensing and electromagnetic spectrum, types of sensors, image interpretation-visual and digital, map, scale, and projections.

Photogrammetric surveys: Basic concepts, principles, and applications of photogrammetry, concepts, and applications for map preparation.

Drone Surveying: Introduction to drones, comparison of surveying drone and its accuracy; techniques of controlling errors

Laboratory Work

1. Measurement of distances/offsets, and bearings with chain and tape, and compass
2. Levelling to calculate RLs of various points
3. Measurement of vertical and horizontal angles with theodolite
4. Tacheometric Survey and determination of tacheometric constants
5. Surveying with Total Station
6. Survey with GPS and DGPS
7. Introduction to GIS software

Course Learning Objectives (CLO)

The students will be able to:

1. Survey an area under various topography and obstructions using various instruments
2. Prepare the plan or map of the area surveyed.
3. Set out curve and building layout.
4. Analyze and process satellite data and retrieve information

Text Books

1. Anderson and Mikhail. Surveying Theory and Practice, 7th Edition, McGraw Hill Education (2012)
2. Duggal, S.K. Surveying, Vol.I and II, 5th Edition, (2019)
3. Subramanian,R. Surveying and Levelling,Oxford 2nd edition (2012)
4. Venkatramaiah,C.,ATextBookofSurveying,UniversitiesPress(2011)
5. PK Garg, Introduction to Unmanned Aerial Vehicles, New Age International Publishers New Age International Private Limited; First edition (1 October 2020); NEW AGE International Pvt Ltd.

Reference Books

1. Punmia, B.C., Jain, Ashok Kumar and Jain, Arun Kumar, Surveying Vol.I and II, Laxmi publications (2016)
2. Garvit Pandya, Basics of Unmanned Aerial Vehicles: Time to start working on Drone Technology, Notion Press (6 March 2021)
3. Kike Calvo, So You Want to Create Maps Using Drones?,Bluerb publication (2015)

Evaluation Scheme

Sr. No.	Evaluation elements	Weightage (%)
1	MST	25
2	EST	35
3	Sessional: (May include the following) Assignment, Regular Lab assessmentand Quizzes	40